

Module 15) Advance Python Programming

3. Opening and Closing Files

- **Opening files in different modes ('r', 'w', 'a', 'r+', 'w+').**

Opening Files in Different Modes in Python – Theory

In Python, files are opened using the built-in `open()` function. The **mode** specified determines how the file will be accessed: for reading, writing, appending, etc.

Syntax:

```
open(file_name, mode)
```

`file_name`: Name or path of the file.

`mode`: A string that defines the file operation (default is 'r' for reading).

Mode Behavior Details:

'r' – Read Only

- File must exist.
- Used when you only need to read contents.
- Reading from a non-existent file causes an error (`FileNotFoundError`).

'w' – Write Only

- Creates a new file or overwrites an existing one.
- All previous content is **erased**.

'a' – Append

- Adds new content **to the end** of the file.
- Does not erase existing content.
- Creates file if it doesn't exist.

'r+' – Read and Write

- Allows both reading and writing.
- File must exist.
- Does not erase existing data but allows overwriting starting from the top.

'w+' – Write and Read

- Creates a new file or truncates existing file.
- Allows both reading and writing.
- Previous content is lost.

- **Using the open() function to create and access files.**

Using the open() Function to Create and Access Files – Theory

In Python, the open() function is used to **create, open, read, write, and append files**. It is the primary way to interact with files stored on disk.

Syntax:

```
file_object = open(file_name, mode)
```

file_name: The name (or path) of the file.

mode: The mode in which to open the file ('r', 'w', 'a').

Purpose of open() Function:

- **Create a new file** (using modes like 'w', 'a', or 'x')
- **Access and read contents** of an existing file ('r', 'r+')
- **Write or append data** to a file ('w', 'a', 'w+', 'r+')

Steps in File Handling:

1. **Open the file** using open().
2. **Perform operations** (read, write).
3. **Close the file** using .close() or use a with block for automatic closing.

- **Closing files using close().**

In Python, when you open a file using the open() function, it's important to **close it properly** after you're done working with it. This is done using the **close()** method.

Purpose of close()

- Releases the **resources** associated with the file.
- Ensures all **data is written** to disk (especially important when writing).
- Prevents **data corruption** or file locking issues.
- Helps maintain good programming habits and resource management.

Syntax:

```
file_object = open("filename.txt", "mode")
```

```
file_object.close()
```

Key Characteristics:

- Once a file is closed using `close()`, you **cannot perform further operations** on it unless reopened.
- Failing to close a file, especially when writing, might result in **loss of data**.